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뇌-마음-행동 개별연구 보고서

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연구 제목	Interpretability of Electrophysiology Foundation Models in decoding auditory processes

1. 연구의 배경 및 중요성

1) Electrophysiological signals and foundation models

Electrophysiological signals have been a valuable source for linking brain activity with identifiable states ranging from physiological homeostasis, movement, and cognition. Among many signal acquisition techniques, intracranial electroencephalography (iEEG) signals driven from mostly human epilepsy patients, show notably high temporal and spatial resolution. Given that the sampling rate of iEEG signals range of 1,000–3,000 Hz, they contain temporal information with millisecond resolution. (Parvizi & Kastner, 2018) Depth iEEG enables recording neural population activity of the limbic structures, which are deeply integrated inside the brain matter. Other than spatiotemporal resolution, iEEG signals portray clear patterns in some frequency bands, corresponding to a certain cognitive or motor state. To utilize these hallmarks of brain states, several processing techniques such as independent component analysis and spectral processing have been employed in fields of clinical diagnosis and brain-computer-interfaces.

However, the inherent diversity of iEEG signals across subjects and recording conditions poses a significant barrier for task-specific supervised workflows. The scarcity of labels in iEEG data confines the amount of usable dataset and limits generalization in traditional supervised learning approaches. Foundation models, by contrast, which are pretrained by self-supervised objectives, such as to reconstruct a part of their own data, are large scale models capable of capturing representations of the data itself (Bommasani et al., 2021). These pretrained representations can then be adapted to a wide range of downstream tasks with minimal fine-tuning. Large language models although being trained by a next-token prediction objective, induce rich internal representations such as the semantic hierarchy of language. Such models offer an opportunity to leverage unlabeled iEEG data to link latent brain states and support transfer learning across subjects, conditions, and tasks.

2) Interpretability in electrophysiology foundation models

Developing foundation models for electrophysiological signals spark the anticipation, that such representations will enable clarifying the complex nature of brain signals. This expectation however, cannot be met unless the model shows robust performance against non-informative perturbations and yields biological meaningful explanations. In the domain of large language models, interpretability has been extensively explored through attribution-based techniques, including gradient-based saliency methods and attention analyses, which have been shown to identify input tokens that are causally relevant for specific downstream predictions. For instance, prior work demonstrates that when state of the art LLMs perform semantic analysis, attribution scores reliably highlight linguistically meaningful tokens that determine the predicted connotation. Analogously, interpretability in electrophysiology foundation models should move beyond predictive accuracy and aim to reveal whether model attributions align with established neurophysiological principles, such as task-relevant frequency bands, anatomically plausible electrode groups, and temporally localized neural dynamics. Establishing such correspondence is critical for validating that learned representations reflect genuine neural processes and for enabling foundation models to serve not only as powerful predictive tools, but also as mechanistic proxies of brain function.

Therefore, this study aims to seek how interpretable, the state-of-the-art electrophysiological foundation model can be when decoding auditory to linguistic features. Here, interpretability is confined to the scope from how the model computes the prediction in the encoder, to which inputs strongly influence the model's classification.

2. 연구 방법 및 결과

1) Model Structure and Task evaluation

An electrophysiological foundation model, DIVER, was utilized for this interpretation research. (HAN et al., 2025) DIVER requires a 500Hz sampled time series of iEEG signals for each channel. The input is then patched into 10 timepoints without overlap, resulting into a shape of the following;

$$X \in R^{C \times N \times 50}, C = \text{number of channels}, N = \text{number of patches} = 10$$

This input's temporal features are extracted with a 3 layer convolutional network mapped into $X \in R^{C \times N \times d}, d = \text{embedding dimension}$. The temporal features are augmented with patch-wise frequency information from FFT, channel position and modality, and relational values to account for cross channel-time relationships using a transformer. The embeddings retain the shape of $X \in R^{C \times N \times d}$. These embeddings, which we call tokens, are then computed through a 12 layered transformer. The transformer utilizes the multi-head self-attention mechanism, so that different heads can identify various context ranges in the tokens. Self-attention is an algorithm which employs an inner product of its queries and keys which are linear transformation of the tokens, eventually to process which tokens are most considerable to a certain query token (Vaswani et al., 2017). Since the model attends to channel and time variables simultaneously, it produces an attention matrix that consists of approximately 1200 queries and 1200 keys for each head and layer (channel number may differ by subject).

$$A \in R^{H \times \mathrm{CN} \times CN}, H = \text{number of heads}$$

Through the values of the attention matrix, we can implicitly see which tokens are salient for the model to perform in its given task. The feature embeddings extracted from the last layer of the transformer are linearly probed to classify binary labels. The downstream task is from Neuroprobe, which consists of iEEG recordings from 10 human subjects performing a naturalistic movie viewing task (Zahorodnii et al., 2025). DIVER shows state of the art performance in this benchmark, when evaluated in the same subject same movie setting. Therefore, we can say DIVER can transfer its learned representations within sessions. Whether its learned weights can transfer between trials and subjects is not yet tested.

Among many naturalistic features, this research confines the interpretation of the foundation model into three tasks; volume, speech, and word head position. Volume task requires classifying whether the average root mean squared watts of the audio were high or low. Speech task requires classifying whether there was speech and word head position task refers to classifying the left/right of the word's dependency in tree head. Notably, word head position is a semantic feature that requires memory in earlier words and integration in the incoming words over time. This scope of tasks spans simple auditory features to complex semantic features, enabling a concise comparison across different levels of the brain's processing hierarchy.

The following results track the model's interpretability in two complementary perspectives: (i) task dependent changes in the attention matrix within subjects and (ii) the importance of input signals to validate whether the model relies on appropriate neural inputs.

2) Attention rollout for interpreting attention weights

The large dimension that the attention matrices take up, needed to be extracted into a smaller dimension for setting intuitive grounds of how the model picks up salient values. Figure1(a) refers to the rank of attention matrices by layer. The corresponding plot, consisting of cross subject and trial matrices, indicates that the rank of each attention matrix by layer is divergent. However, simply attending the attention values cannot demonstrate which tokens are salient due to the residual network and recursive multiplication throughout layers. Therefore, we here use the attention rollout mechanism which accounts for all layers and the

residual network between each transformer blocks (Abnar, Zuidema 2020). The following equation is employed to extract the attention rollout matrices. Every update of the attention rollout matrix entails a row normalization for restricting the sum of attention weights to 1 for each query.

$$\hat{A}_{\mathrm{L}} = (A_L + I)\hat{A}_{L-1}$$

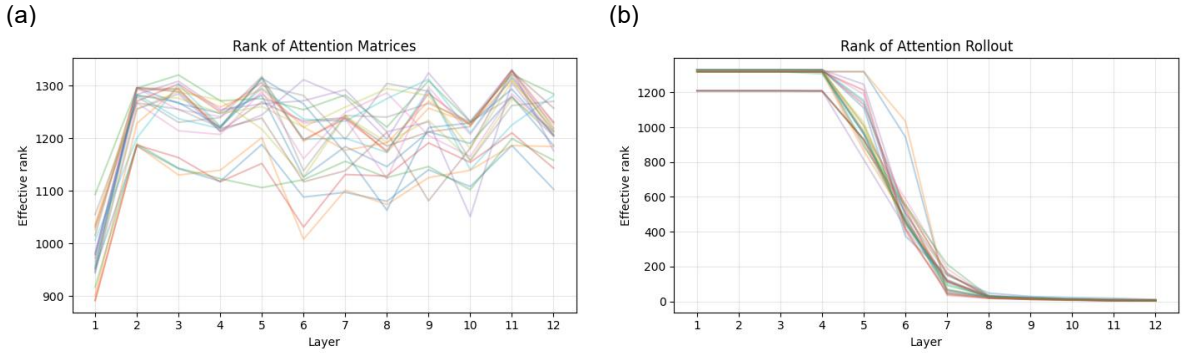

Figure 1 (a) Rank of attention matrices across layers**, (b)** rank of attention rollout matrices across layers, both plots have samples across different subjects and trials

As Figure1(b) shows, the attention rollout matrix shows a rank collapse at layer 7 and 8, resulting in a rank 1 matrix which is capable of maximum rank CN. This implies that for every query, the attention weights are equivalent in the final computation. Therefore, we can use any arbitrary row of attention matrix which has the dimension of the following $\hat{A}_{L_i} \in R^{CN}, i \in 1,2,\ldots,CN$, where every query attends to the same pattern.

3) Attention rollout patterns across and within sessions

Before capturing the salient attention values for each task, comparison of attention values within and between sessions was conducted using a two-sample multivariate comparison test. To assess session-level similarity of attention patterns, we first applied the two-sample Energy distance test to vectorized attention rollout maps $\hat{A}_{L_i} \in \mathrm{R}^{CN}$. Although the Energy test detected statistically significant differences across most session pairs, including within-trial comparisons, this result likely reflects the extreme sensitivity of the test in high-dimensional settings rather than substantive changes in attention structure. Accordingly, we focused on reproducible *salient* attention patterns by extracting the top 10% highest-attention channel–time patches. Subject intersection was infeasible due to subject specific electrode configurations. The left column of Figure 2 depicts the number of intersected channel-time patches within and between sessions.

The right column of Figure 2 shows the attention maps highlighted by the overlap within a subject. Volume, a low-level acoustic feature, engages several regions including the medial frontal gyrus, superior frontal gyrus and the amygdala. The amygdala is also known to process acoustic information when relevant to discriminate acoustic emotion (Pannese et al., 2016). Such prior results implicate the role of the amygdala's signal in processing the amplitude of the volume. The speech task's attention map reveals, the initial time tokens in the superior frontal gyrus (SFG) are attended the most. Considering that the speech occurred at the baseline 0.0s of the sample and that the SFG is known to be activated when hearing human voices, this result is in line with biological relevance. In contrast with the volume task, speech task entails fewer overlap of regions, which reflect the task's characteristic, requiring fewer selective regions for processing. In the word-head-position task, the time range of salient patches is longer than the other tasks. Considering that this task requires working memory of the previous and current words, processing of later values is plausible. The channel token that

is highlighted in a lengthy time range corresponds to the rectus gyrus. The region's function is not known clearly, which impedes interpretation for now.

Overall, the attention map across the channel and time patches reveals temporal and anatomical resemblance to the naturalistic responses of the brain.

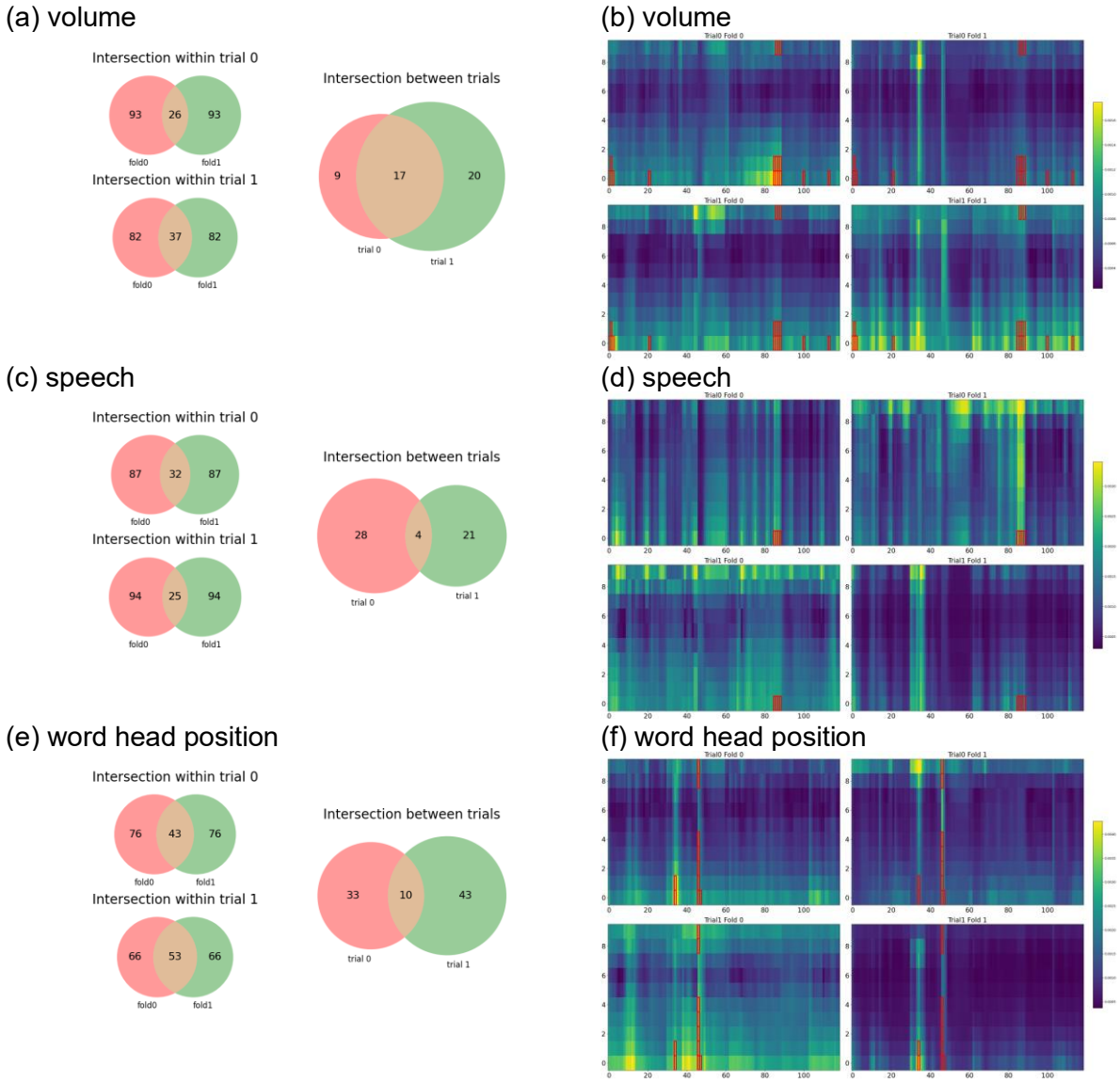

Figure 2 (a), (c), (e) each corresponds to the overlap of channel-time patches which have the maximum 10% attention values. **(b), (d), (f)** each corresponds to the attention map where x axis refers to the channel index and the y axis refers to the time index. The brighter colored grids refer to higher attention rollout values and the red borders indicate the overlapped channel-time patches within and between sessions. The first, second, third row each indicates volume, speech, and word head position task.

Task	
Volume	[0.0~0.1s] Hippocampus, [0.0~0.1s] Middle Frontal Gyrus, [0.0~0.1s] Frontal Inferior Triangular Gyrus, [0.0~0.1s] Cingular Anterior Gyrus, Sulcus, [0.0~0.2s] Amygdala, [0.0~0.2s, 0.9~1.0s] Superior Frontal Gyrus
Speech	[0.0~0.1s] Superior Frontal Gyrus
Word Head Position	[0.0~0.2s] Hippocampus, [0.0~0.5s, 0.8~1.0s] Rectus Gyrus

Table 1 Channel and time patches which are overlapped within and between sessions

4) Patch importance through perturbation methods

To determine how each patch's frequency information influences the model's performance, a frequency perturbation analysis was performed. Within each time patch, a Gaussian kernel was constructed based on pairwise time-point differences and was applied to compute a weighted temporal average, such that nearby time points contributed more strongly than distant ones. Increasing the standard deviation meant having smaller frequency restrictions on the low pass filter. This perturbation acted as a low-pass filter within the patch, applied to every patch in a certain channel. For both kernels of standard deviation 3 and 10, signals with frequency of approximately 22Hz and 6Hz were silenced. However performance changes induced by perturbations were extremely small ($|\Delta| < 10^{-3}$) across all conditions, indicating strong robustness. While some perturbation indices showed larger mean effects, these were highly trial-dependent and lacked consistent directionality across folds. Such results imply that the model does not take account of that certain frequency bands as much.

To quantify the contribution of each channel-time patch to task performance, a Monte Carlo perturbation method, which randomly selects subsets of channel-time patches for each run, was used. For each iteration, a binary mask was sampled over channel-time patches, allowing 10% of the total patches to be perturbed. The perturbation method was to zero out patches, removing both temporal and frequency structure. Model performance (AUROC) under each perturbed input was compared to the preserved baseline, and the resulting performance degradation was attributed back to the perturbed indices. r corresponds to each iteration ranging from 1 to R and F refers to the model performance statistic.

$$M^{(r)} \in \{0,1\}^{C\times \mathrm{N}}$$

$$\Delta^{(\mathrm{r})} = F_0 - F^{(r)}$$

$$Imp(c,n) = \frac{1}{R}\Sigma_{r=1}^{R}\Delta^{(r)}(1 - M_{c,n}^{(r)})$$

Importance maps showed strong within-trial reproducibility across speech and word head position task (Spearman ρ = 0.80, and 0.495), whereas between-trial comparisons yielded near-zero correlations (ρ range −0.02 to 0.09). A permutation test showed a consistent trend toward greater within-trial similarity (p ≈ 0.06 for all tasks), although statistical power was limited by the small number of session pairs. These results imply that in the same subject, different contexts induce the model to use different channel-time patches as relevant sources.

Despite big variances between trials, channel-time patches which had bigger importance scores bigger than the 2 standard deviations above the mean for each session were intersected. For the speech task, the gyrus rectus in the 7th patch showed overlapped salience while for the word head position task, the frontal inferior sulcus in the 4th patch showed overlapped salience.

3. 연구의 의미

1) Spatial and temporal relevance in attention weights and input importance scores

The attention rollout weights derived for each task within trials, showed overlaps in regions that are known for its function in high order processing and sensitivity to auditory processes. Notably the attention weights revealed task specific characteristics in spatial and temporal dimensions. The word head position task retained high attention values in a single channel for a longer time range than simple auditory decoding tasks. The volume task which engages larger regions than the speech task for processing in the brain, also entails more channels in big attention weights. Attention rollout weights showed significant variances between trial and session, while input importance score showed high correlation within trials. This result implies that the model takes account of similar channel and time patches in a familiar context, while this does not correspond to the computation of attention weights. Since attention is a

bidirectional computation between all possible variables, predicting its salience will require methods that take account of the computation characterstics more sensitively.

2) The influence of depth in rank of attention weights

The collapse of rank in attention rollout matrices is parallel to prior studies reporting the inevitable rank collapse in transformers (Dong et al., 2021). Despite taking account into the residual layer, the attention rollout shows a dramatic collapse of rank 1, with the biggest singular value approximating to 1, while the others are insignificant values. Elaborating on why such phenomena occurs and if the middle layers compute better representations of brain signals is not yet known. Considering the pretraining objective which is to reconstruct the masked patches, the final layers may be tuned to perform successful local interpolations rather than meaningful global representation. How the final attention rollout matrix converges to such rank 1 matrix should be further elaborated. Comparing the model's performance across layers of transformers will enable the understanding of the depth of the transformers.

3) Perturbation analysis and safety of electrophysiology foundation models

Overall, understanding why and how AI models make decisions is a critical process to apply such models in real life examples. This study revealed that low pass filtering 6Hz and 22Hz don't significantly influence the model's performance. By varying the filter for each input, which frequency signals are critical for the model's processing may be understood thoroughly. To destruct the time signal, a blank out method (zeroing out) was used, because temporal averaging through high variance gaussian filters didn't induce meaningful changes in the model's behavior. Zeroing out patches showed that important input patches were dependent on the context. By aligning input scores and the naturalistic feature data, further understanding in the divergent computations of the foundation model will be possible. Lastly, this perturbation interpretability designs allow which components are significant components for EEG dynamics to be understood and how the brain and cognitive functions are linked. Such research will accelerate building not only predictably accurate but safer and reliable foundation models.

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